

Rapid Cross-Domain Transfer in Wearable Systems via Unsupervised Latent Space Prediction

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Abstract

The deployment of wearable gesture recognition systems is hindered by the prohibitive cost of labeling sensor data for each new user and domain. While Self-Supervised Learning (SSL) has revolutionized learning systems for dense modalities like images and video, its application to sparse, heterogeneous sensor streams remains underexplored. We present a hardware-agnostic generalized kinematic encoder based on the Joint Embedding Predictive Architecture (JEPA). Unlike approaches that rely on unscalable contrastive learning and heuristic-based approaches such as teacher-student exponential moving average (EMA) weight updates, we use Sketched Isotropic Gaussian Regularization (SIGReg), which theoretically prevents representation collapse and enforces an information-maximizing distribution in the latent space. We validate our framework on Cross-Domain Transfer tasks by pretraining the encoder on unlabeled high-complexity static primitives like *Kathakkali Mudras* and *Kuji-in* (Ninja Hand Seals), using datasets collected on our custom-designed sensorized wearable platform. We demonstrate that pretraining on biomechanically complex primitives enables superior transferability, achieving 42.35% accuracy on American Sign Language Recognition with only 1% of the training data, a 17.22% improvement over a fully supervised baseline.

1 Introduction

Approximately 70 million people worldwide are deaf and mute, while over 300 million individuals experience significant hearing impairments [World Federation of the Deaf, 2023; World Health Organization, 2024]. Despite its effectiveness within the deaf community, Sign Language poses a significant communication barrier when interacting with non-sign language users, particularly in educational, social, and professional environments. This communication gap has motivated researchers to develop automated sign language translation systems. Such systems aim to facilitate interaction between individuals with speech or hearing impairments

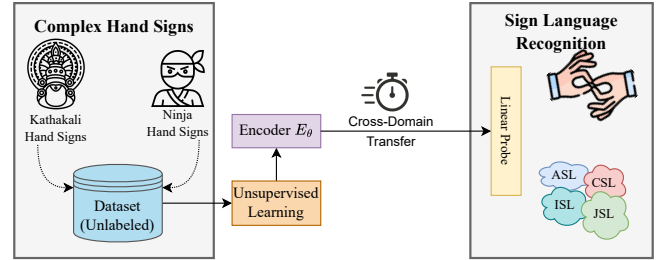


Figure 1: Pretraining an unsupervised encoder on complex hand sign data helps rapid cross-domain transfer learning to applications such as Sign Language Recognition.

and those unfamiliar with sign language. However, the prohibitive costs of labeling sensor data continue to impede the development and reliable deployment of these gesture recognition systems. This gives rise to a need to develop systems capable of learning universal, calibration-free representations that enable rapid cross-domain transfer across domains and languages.

While computer vision (CV) powers most existing solutions, it is inherently prone to occlusion, privacy, and environmental lighting-related issues [Aditya *et al.*, 2024; Almjally and Almkadi, 2025; Shabana Parveen *et al.*, 2024; P *et al.*, 2024], emphasizing the need to develop robust, rapidly adaptable, and wearable sensorized gesture recognition systems. The scarcity of labeled data drove the development of self-supervised learning (SSL) methods [Sohl-Dickstein *et al.*, 2015; Zbontar *et al.*, 2021], which are marred by problems like prohibitive negative sampling for contrastive models [Chen *et al.*, 2020], inefficient utilization of computational resources when modeling high-frequency noise [He *et al.*, 2022], and unscalable heuristics. The Joint Embedding Predictive Architecture (JEPA) [LeCun and Courant, 2022] solves both aforementioned issues as it is: (a) Non-contrastive; it does not require negative samples, and (b) Latent predictive; it predicts latent representations instead of reconstructing the raw data. However, its application to sparse, heterogeneous sensor streams remains unexplored.

We adapt the JEPA architecture for kinematic modeling of the hand-articulation manifold. We replace the heuristic-based teacher-student EMA weight updates in the original JEPA with Sketched Isotropic Gaussian regularization (SI-

GReg) [Balestriero and LeCun, 2025], which provides theoretical guarantees against representation collapse. Our primary objective is to demonstrate the rapid cross-domain transfer learning capability of our pretrained encoder on datasets collected entirely on our custom-designed hardware. For pretraining, we use a dataset of unlabeled sensor data of *Kathakali Mudras* [Thampuran, 1892] and *Ninja Kuji-in* [Vandeyck, 2017] hand signs. Due to their highly diverse hand articulations, they help the encoder capture the entire hand kinematics manifold. The pretrained model may then be used as a feature extractor for cross-domain tasks, such as Sign Language Recognition (SLR). For evaluation, we focus on a linear probe trained on American Sign Language (ASL) due to its widespread use, and compare against a fully supervised baseline. Our method achieves a 17.22% accuracy lead over supervised baselines with just 1% labeled data, demonstrating the first application of JEPA to sparse, heterogeneous sensor streams. Additionally, our method achieves a doubled first-epoch accuracy, facilitating universal manifold learning through unsupervised pretraining on complex primitives.

The main contributions of our work are summarized below:

1. First application of JEPA with SIGReg regularization to sparse, heterogeneous sensor streams
2. Hardware agnostic tokenization and masking strategy for robust kinematic feature extraction
3. Empirical proof of superior manifold learning and label efficiency, accelerating and stabilizing cross-domain transfer learning

The remainder of this paper is organized as follows: Section 2 reviews the related work. Section 3 describes the proposed methodology, followed by Section 4, which outlines the experimental setup and implementation. Section 5 presents and discusses the results. Finally, Section 6 concludes the paper and highlights directions for future research.

2 Related Work

2.1 Gesture Recognition for Sign Language Translation

SLR systems are broadly categorized into vision-based and sensor-based approaches. Vision-based methods commonly rely on hand tracking and deep learning; for example, a Mediapipe-based hand tracking system integrated with a CNN achieved 99.12% accuracy on American Sign Language (ASL) datasets through efficient preprocessing of hand movements [Aditya *et al.*, 2024].

Sensor-based approaches use wearable devices, such as gloves equipped with flex sensors, accelerometers, and inertial measurement units, to capture detailed hand kinematics. A smart glove-based Arabic SLR system using flex sensors and CNNs achieved 95% accuracy [Ahmed *et al.*, 2023], while a wearable glove system for Korean Sign Language learning employed LSTM networks and attained 85% accuracy [Baek, 2023]. More recently, the CaptAinGlove system combined capacitive and inertial sensors with convolutional neural networks and hierarchical multimodal fusion, achieving 80% training accuracy and 67% testing accuracy in real-time on-device recognition [Bello *et al.*, 2023].

Despite these advances, dynamic gesture recognition remains challenging. A Japanese fingerspelling recognition system using lightweight sensor gloves reported approximately 70% accuracy, underscoring the difficulty of classifying temporal hand movements [Tsuchiya *et al.*, 2020]. Hybrid approaches combining wearable sensors with vision-based techniques have also been explored; for instance, a data glove system using a multi-layer perceptron achieved high translation accuracy with reduced computational complexity [Menon *et al.*, 2021].

Localized SLR has gained attention, with frameworks developed for region-specific languages such as Hindi Sign Language, highlighting the limitations of systems primarily designed for ASL [Ifham *et al.*, 2023]. Beyond sign language translation, wearable glove systems have also been applied to human-computer interaction by capturing finger flexion and hand orientation using flex sensors and accelerometers [Cabrera *et al.*, 2014].

2.2 Environmental Constraints of Computer Vision (CV)

CV-based SLR systems face significant challenges in deployment despite their maturity. Fundamental constraints include line-of-sight requirements and occlusion, where self-occlusion of signs or external obstacles leads to catastrophic failure [Gao *et al.*, 2024]. Performance further fluctuates under dynamic environmental conditions such as low-light or high-glare settings. Furthermore, the constant use of cameras introduces privacy concerns, limiting adoption in sensitive environments.

Our sensorized wearable setup aims to overcome these challenges. By capturing biomechanical intent via IMUs and flex sensors, the system is robust to environmental conditions. Wearables bypass occlusion entirely, providing reliable tracking [Chen *et al.*, 2025]. This modality is inherently privacy-preserving and computationally efficient, as processing sensor data streams is well-suited for power-constrained edge devices, granting proper mobility and portability [Amin *et al.*, 2022].

2.3 Limitations of Contrastive and Generative SSL

SSL provides a mechanism to utilize unlabeled sensor data. However, standard paradigms face significant challenges in the context of assistive technologies.

Contrastive Learning Data augmentations in frameworks like [Chen *et al.*, 2020] may be semantically destructive for SLR applications. Minor shifts in the distribution of sensor signals, which contrastive models treat as noise, often carry semantic meaning, leading to false negatives in recognition [Gan *et al.*, 2023].

Masked Autoencoders (MAE) Generative SSL reconstructs raw signals that contain high-frequency noise in the context of wearable sensors. Masked autoencoders [He *et al.*, 2022] for SSL on such data introduce significant computational overhead by modeling noise rather than the underlying biomechanical intent [Cao *et al.*, 2023]. For embedded devices, these inefficiencies limit practical utility.

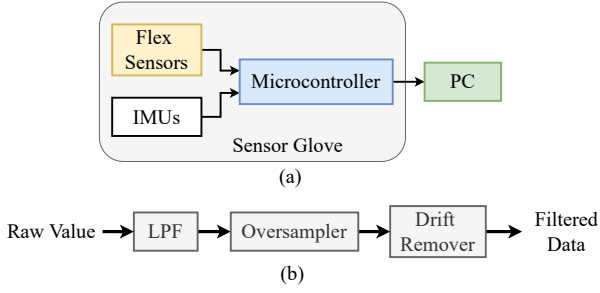


Figure 2: (a) Sensor glove architecture, (b) Filter applied on raw data to remove noise and drift.

2.4 Latent Predictive Modeling with JEPA and SIGReg

To prioritize semantic intent over raw signal noise, we utilize JEPA [LeCun and Courant, 2022]. JEPA represents a fundamental shift in the self-supervised learning landscape, pivoting from the computational overhead of generative reconstruction and the augmentation-dependence of contrastive methods toward the acquisition of hierarchical world models [Assran *et al.*, 2023; Assran *et al.*, 2025]. By facilitating the capture of high-level semantic structures through latent prediction, this architecture establishes a robust framework for learning representations that are invariant to low-level signal fluctuations. By predicting latent representations rather than reconstructing input-level sensor values, JEPA focuses on the abstract dynamics of movement. To address Representation Collapse, we incorporate Sketched Isotropic Gaussian Regularization (SIGReg) [Balestrieri and LeCun, 2025]. Unlike heuristic-based solutions, SIGReg provides a theoretically grounded method to ensure the latent space remains information-rich and expressive across diverse populations. This framework provides a robust, low-latency interface that remains resilient to the hardware noise and calibration issues inherent to wearable SLR.

3 Methodology

We aim to train a module to form emergent intelligence about different gestures from sensor readings alone through unsupervised learning, training the model to interpret data without labels. JEPA [LeCun and Courant, 2022] provides a solution based on non-contrastive energy-based models (EBMs) to perform this kind of training. The upcoming sections discuss the construction of the JEPA for unsupervised understanding of gestures from sensor readings.

3.1 Data Acquisition Pipeline

For this work, we developed a custom wearable setup for hand-gesture recognition, integrating multiple sensors, including flex sensors (B07QC8S3Z8) and an Inertial Measurement Unit, *aka* IMU (LSM6DS3). Two such gloves were constructed for the left and right hands, forming an enantiomorphic pair, which ensures symmetric sensing across both hands. The data flow follows a structured pipeline: sensor measurements are first acquired by the glove, transmitted to an onboard microcontroller, and finally streamed to a host

PC for data collection, filtering, and storage as shown in Figure 2(a). The raw signals are processed sequentially through three filtering stages, as shown in Figure 2(b):

1. A low-pass filter (LPF) to suppress high-frequency noise.
2. Moving-average (oversampling) filter to smooth temporal fluctuations.
3. Drift-removal filter to compensate for slow baseline drift and correct deviations from the mean hand posture.

To maintain data consistency and reduce inter-subject variability, all gesture data is collected from a single individual. This ensures that all Kathakali Mudras, Ninja Kuji-in signs, and ASL gestures are performed by the same subject, enabling controlled and fair comparison across sign systems. Rather than introducing subject-level variance, this design choice isolates the model’s ability to learn transferable representations across structurally diverse gesture domains.

3.2 Tokenizing Sensor Readings

For an n sensor setup, individual sensor readings are transformed to tokens, which are d -dimensional vectors, using per-sensor linear transforms using

$$u_k = \mathbf{W}_s^{(k)}(v_k) \quad \forall k = 1, 2, \dots, n, \quad (1)$$

where $u_k \in \mathbb{R}^d$ is the k^{th} sensor token, $v_k \in \mathbb{R}$ is the k^{th} sensor reading and $\mathbf{W}_s^{(k)}$ is the linear layer for the k^{th} sensor.

To differentiate between sensors in subsequent transformer modules in the pipeline, we add a learnable positional (sensor) embedding to each token u_k . The set of tokens after applying the positional embeddings is denoted as

$$T = \{\tau_1, \tau_2, \dots, \tau_n\} \quad (2a)$$

$$\tau_k = u_k + \text{pe}_k, \quad (2b)$$

where τ_k is the k^{th} positionalized token and pe_k is the sensor (positional) embedding for the k^{th} sensor token.

3.3 Sensor Masking

With the aim to learn latents that are robust to sensor malfunction, we introduce masking of sensors. To mask sensor information, we employ a random masking technique on the tokens in T , where p_{mask} ratio of tokens are dropped. The extra compute overhead in converting sensor readings to tokens, sequentializing them, and dropping tokens is intentional as it provides a theoretically more robust masking paradigm than naively setting sensor readings to zero to mask them. The set of tokens after masking is given by

$$T_{\text{mask}} = \text{Mask}(T, p_{\text{mask}}) \quad (3)$$

3.4 Training the JEPA

JEPA [LeCun and Courant, 2022] aims to train an encoder, E_θ , using two different views of the same data, the context x and the target y , such that the encoder maximizes the information about y in $E_\theta(x)$. A relatively weak predictor, D_ϕ , acts as a bottleneck, pushing the encoder to extract the most relevant information in order to predict the latent representation of y , $E_\theta(y)$. Representation collapse is a phenomenon in which the encoder converges to the trivial solution of outputting a constant C for every input. Popular

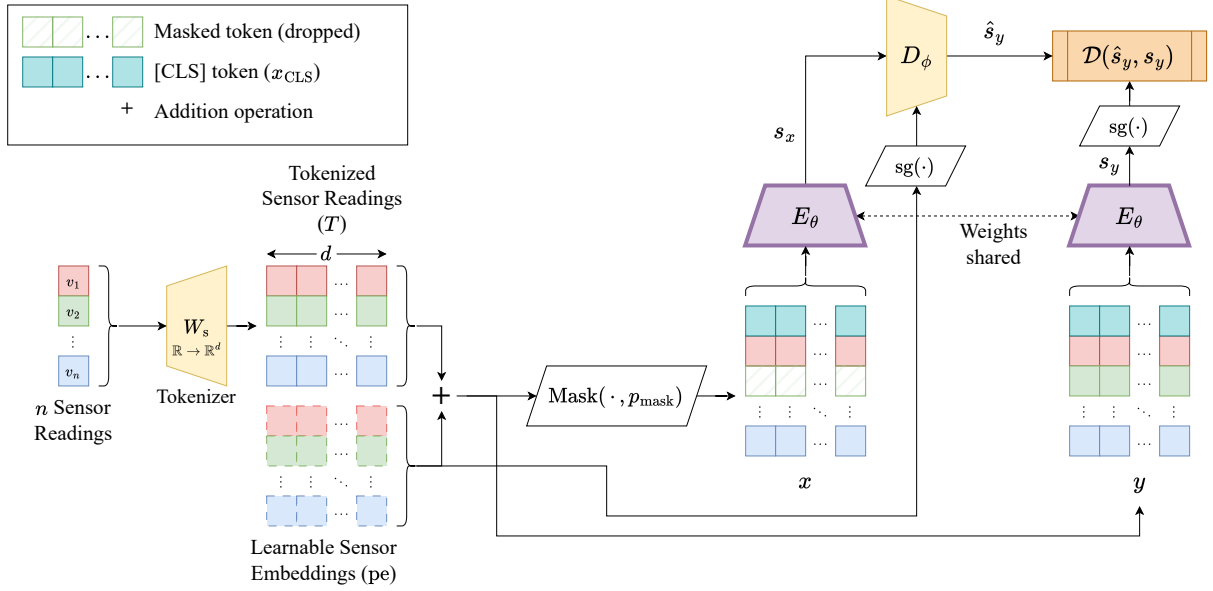


Figure 3: **The JEPA Pretraining Pipeline.** A linear tokenizer network first converts each sensor stream reading to a d -dimensional token. Learned positional embeddings pe are added to the tokens to encode sensor-specific information, to construct the positionalized tokens T . The context x for JEPA is constructed from T by dropping tokens as the masking mechanism, and prepending a [CLS] token. The target y is constructed by prepending the [CLS] token to T . The encoder E_θ encodes x and y to s_x and s_y respectively. The predictor D_ϕ reconstructs the target encoding as $\hat{s}_y$, conditioned on pe .

methods like I-JEPA [Assran *et al.*, 2023] and V-JEPA [Assran *et al.*, 2025] define the target encoder weights by an exponential moving average (EMA) of the context encoder weights, which requires extensive hyperparameter tuning and introduces instability during training. In contrast, we use SIGReg [Balestrierio and LeCun, 2025], which provides theoretical guarantees to prevent collapse.

We prepend a learnable [CLS] token $\mathbf{x}_{\text{CLS}} \in \mathbb{R}^d$ to the T and T_{mask} , similar to the BERT [Devlin *et al.*, 2019], to construct the context x and target y given by

$$x = \mathbf{x}_{\text{CLS}} \oplus T_{\text{mask}} \quad (4a)$$

$$y = \mathbf{x}_{\text{CLS}} \oplus T, \quad (4b)$$

where $\oplus$ denotes the concatenation operation.

As given in Fig. 3, we minimize the JEPA latent prediction loss given by

$$\mathcal{L}_{\text{JEPA}} = \mathcal{D}(D_\phi(E_\theta(x), \text{sg}(pe)), \text{sg}(E_\theta(y))), \quad (5)$$

where $\mathcal{D}$ is a divergence metric, typically the L2 loss, and $\text{sg}(\cdot)$ is the stop gradient operator, which stops gradient flow through the target encoder. The predictor D_ϕ takes in the encoder output and the learnable sensor embeddings pe from (2b) and reconstructs the unmasked sensor tokens. We stop gradients to pe from the predictor circuit to stabilize training.

Representation collapse is prevented by the SIGReg regularization loss, given by

$$\mathcal{L}_{\text{SIGReg}}(\mathbf{X}) = \mathbb{E}_{\mathbf{w} \sim \mathcal{S}^d} [\text{EP}(\mathbf{w}^\top \mathbf{X}, \mathcal{N}(\mathbf{0}, I_d))], \quad (6)$$

where $\mathbf{X} \in \mathbb{R}^{M \times d}$ is a set of $M > 1$, d -dimensional latent encodings, $\mathbf{w}$ is a random direction uniformly sampled from

the d -dimensional unit sphere $\mathbb{S}^d$, EP is the Epps-Pulley test statistic. Note that $\mathcal{L}_{\text{SIGReg}}$ is calculated only on the output [CLS] tokens from the encoder.

Thus, we combine the losses from (5) and (6) to construct the loss function

$$\mathcal{L} = (1 - \lambda)\mathcal{L}_{\text{JEPA}} + \lambda\mathcal{L}_{\text{SIGReg}}(E_\theta(x)), \quad (7)$$

where $\lambda \in \mathbb{R}$ is a hyperparameter.

4 Experimental Implementation

4.1 Data Efficiency and Transferability

To evaluate the data efficiency and transferability of our self-supervised pretrained encoder, we compare linear probing results on an American Sign Language (ASL) dataset collected on the same hardware as the pre-training data. Figure 4 shows the two networks with their respective frozen and trainable modules. We compare validation top-1 accuracy on training both networks on differently sized subsets of the ASL dataset and compare their performance in Section 5.1.

4.2 Visualization of JEPA Latents

To visualize the latent space learnt by the JEPA encoder, we analyze t-SNE and PCA plots of the encoded sign-language sensor readings, colored by the corresponding letters.

4.3 Implementation Details

Our encoder E_θ and predictor D_ϕ (only for pre-training) are transformers [Vaswani *et al.*, 2017] with $d = 128$ and $h = 8$ heads. The sensor embeddings are initialized from $\mathcal{N}(\mathbf{0}, 0.02I_d)$ to preserve the training signals from the sensor

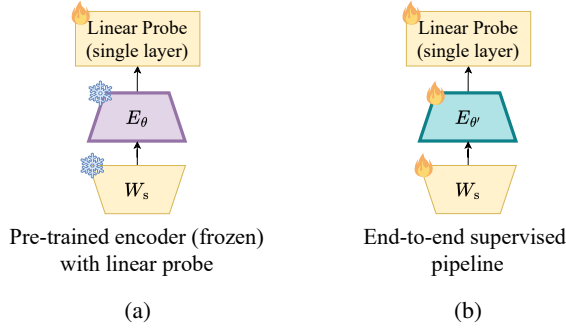


Figure 4: Architectures for comparison. (a) The pre-trained encoder and tokenizer for linear probing, (b) End-to-end fully supervised network.

Table 1: Hyperparameter configurations for JEPA pre-training and downstream ASL linear probing.

Category	Hyperparameter	Notation	Value
Architecture	Embedding Dimension	d	128
	Transformer Layers	-	4
	Attention Heads	-	8
	KV Heads	-	2
	Activation	-	SwiGLU
JEPA Setup	Masking Ratio	p_{mask}	0.33 to 0.5
	SIGReg Weight	λ	0.001 to 0.5
	Context-Target Weights	θ	Shared
Optimization	Optimizer	-	AdamW
	E_θ Learning Rate	-	1×10^{-4}
	D_ϕ Learning Rate	-	1.5×10^{-4}
	W_s Learning Rate	-	5×10^{-5}
	Probe Learning Rate	-	1×10^{-3}
	Batch Size	-	8192
Training	Pre-training Epochs	-	300
	Linear Probe Epochs	-	15

readings in the tokens τ_k . We implement grouped-query attention (GQA) [Ainslie *et al.*, 2023] with 2 KV heads. For pre-training the JEPA we schedule the SIGReg λ parameter from 0.001 $\rightarrow$ 0.5, and the masking ratio p_{mask} from 0.33 $\rightarrow$ 0.5, through 300 epochs of training. Pre-training is conducted on the *Kathakali Mudra* (28 signs) and *Ninja Kuji-in* (12 signs) datasets. Downstream training on ASL data (26 alphabets) is performed for 15 epochs. All training was performed on an NVIDIA RTX A4500 GPU. Table 1 gives a summary of all other hyperparameters used in our experiments.

5 Results and Discussions

5.1 Data Efficiency and Transferability

The results of training a linear probe in the pre-trained architecture and an end-to-end fully supervised network on differently sized subsets of the ASL dataset are provided in Table 2, shown graphically in Figure 6. The data efficiency of the JEPA pre-trained encoder is more pronounced in the low-data regime, where it achieves 42.35% accuracy, much higher

than that of the fully-supervised one (25.13%), when trained on just 1% of the entire dataset. This indicates that unsupervised pre-training on data with complex gestures such as *Kathakali Mudras* and *Ninja Kuji-in* helps the encoder capture the universal manifold of hand gestures, thus generalizing to domains like SLR with minimal supervision, whereas the supervised model fails to learn a coherent mapping from scratch when examples are limited.

A crossover is observed after training on more than 50% of the data, where the fully-supervised network specializes its entire backbone, outweighing JEPA’s pretrained knowledge, reaching 92.92% accuracy on the entire dataset, beating JEPA’s competitive 87.30%. In real scenarios where labeling 100% of the data is unfeasible, JEPA provides the optimal labeling effort to performance trade-off. Despite a slightly lower accuracy at the end of training, the JEPA encoder demonstrates a significant “cold-start” advantage at the early stages of downstream adaptation on 100% of the ASL data (See Figure 4). Specifically, the JEPA encoder starts with an epoch 1 accuracy of 29.45%, more than double the 14.01% baseline achieved by the fully-supervised model, further corroborating that the JEPA encoder has learned a robust prior manifold for gesture recognition. Analysis of the training dynamics in low data regimes (1%-10% data) highlights the superior training stability of training on the pre-trained features. While the JEPA linear probe shows smooth and monotonic convergence in Figure 5a, the fully supervised baseline exhibits significant volatility, as seen in Figure 5b.

5.2 JEPA Latent Space

We evaluate the qualitative and quantitative characteristics of the learned representations to examine the structural trade-offs between task-specific specialization and generalized kinematic modeling.

Analysis of the Learned Manifolds

The latent space geometry on the ASL dataset reveals distinct architectural behaviors. As shown in Table 3, the supervised model achieves a superior Silhouette Score of 0.3456. This high cluster cohesion is a direct consequence of the supervised objective, which explicitly optimizes for class discriminability by enforcing isolated, high-density regions for specific labels.

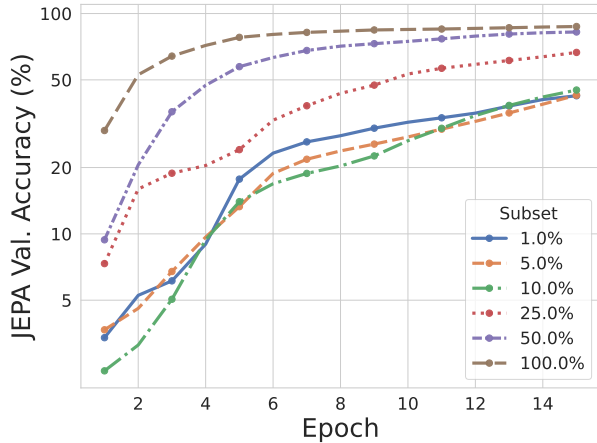
However, the JEPA encoder exhibits significantly higher feature expressivity, with a latent variance of 0.7068 compared to 0.5279 for the supervised baseline. This increase in variance indicates that the JEPA encoder utilizes the entirety of the latent space to encode sensor data inputs, as enforced by the SIGReg regularization, thus mitigating the risk of representation collapse.

Topological Distribution

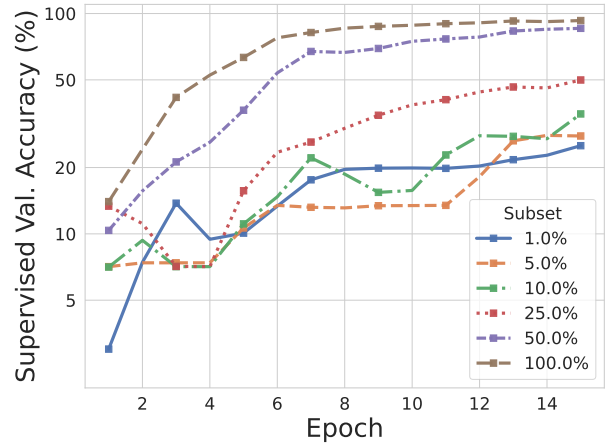
Visualization via t-SNE and PCA clarifies these observations. The JEPA PCA projections in Figure 7a exhibit a distribution with less clustering around the origin and latents that spread out in the latent space. Conversely, the supervised PCA in Figure 7b displays significant feature crowding at the origin, indicative of representation collapse, where the model has only learned a manifold with just enough variance to help its downstream classification task.

Table 2: Comparison of Classification Accuracy across varying fractions of the ASL Training Set. The pre-trained JEPA encoder (frozen) significantly outperforms the fully supervised baseline in the low-data regime.

Training Data Fraction	Fully Supervised (End-to-End)	JEPA (Pre-trained Frozen)	Improvement (Δ)
1.0%	25.13%	42.35%	+17.22%
5.0%	27.81%	42.57%	+14.76%
10.0%	35.02%	44.95%	+9.93%
25.0%	49.93%	66.60%	+16.67%
50.0%	85.66%	82.46%	-3.20%
100.0%	92.92%	87.30%	-5.62%



(a)



(b)

Figure 5: Training curves for different subsets of ASL data, for the networks (a) Pretrained JEPA encoder with linear probe (b) End-to-end fully supervised network

Local clustering analysis via t-SNE reveals that while the supervised objective enforces high intra-cluster density and inter-cluster sparsity, resulting in disjoint decision boundaries, the JEPA manifold preserves a higher degree of semantic continuity. This suggests that the JEPA encoder captures the inductive bias of the underlying hand-articulation manifold, maintaining the semantic relationships between similar gestures, such as those sharing hand orientations or bending of the same finger. In contrast, the supervised encoder learns to discriminate clusters just to minimize the classification loss, resulting in arbitrary partitions which disregard the topological relations between hand-articulations.

Implications for Downstream Tasks

These findings provide a qualitative explanation for the observed data-efficiency and adaptability characteristics of the JEPA encoder. Learning a high variance manifold, via SIGReg regularization, helps it act as a robust world model. This enables superior performance on data-scarce regimes, as the downstream network needs only to learn to map an expressive prior manifold to labels, rather than learn feature extraction from limited samples.

6 Conclusion and Future Work

This work demonstrates that unsupervised latent predictive modeling is a viable and effective strategy for cross-domain

Table 3: Manifold Structural Analysis

Task	Metric	Sup.	JEPA
Pretrain (Kathakali + Ninja)	Silhouette	0.230	0.241
	Variance	0.346	0.513
Transfer (ASL)	Silhouette	0.346	0.204
	Variance	0.528	0.707

transfer in wearable gesture recognition systems. By adapting Joint Embedding Predictive Architectures to sparse, heterogeneous sensor streams and eliminating heuristic stabilization via SIGReg, we show that it is possible to learn an information-rich, non-collapsed kinematic manifold without labeled data. Pre-training on biomechanically complex gesture primitives enables rapid adaptation to downstream sign language recognition tasks, delivering substantial gains in data efficiency and cold-start performance.

Overall, this study establishes that predictive, non-contrastive self-supervised learning can move wearable gesture recognition beyond memorization of voltage patterns toward universal, hardware-resilient representations of human gestures. This paradigm substantially reduces labeling cost, improves deployability across users and domains, and aligns well with the constraints of privacy-preserving,

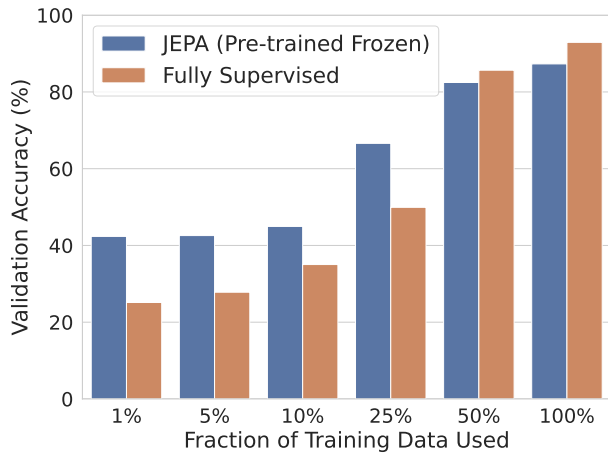


Figure 6: Comparison of Data Efficiency between Pre-trained JEPA and Fully Supervised Learning. The JEPA encoder, pre-trained on Kathakali sensor data, demonstrates significant transferability to ASL gesture recognition, outperforming supervised training in the low-data regime.

energy-efficient assistive technologies. Future work will extend this framework to dynamic temporal modeling and on-device continual adaptation in real-world wearable deployments.

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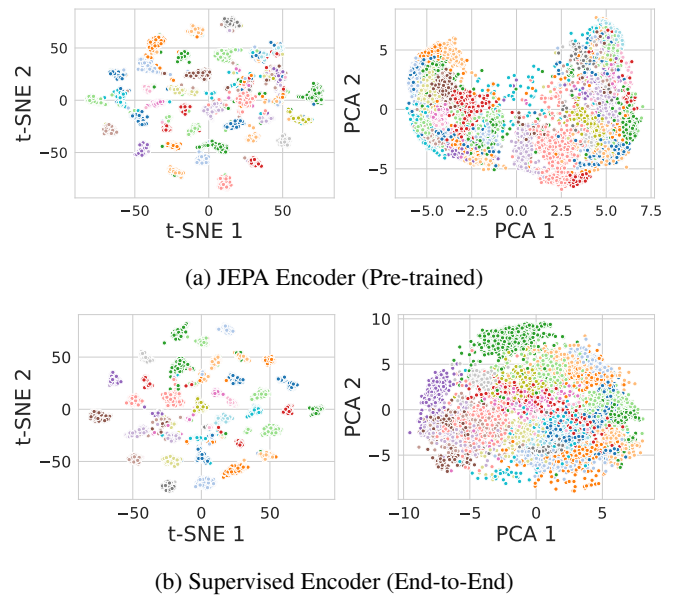


Figure 7: Comparison of latent manifold structural characteristics on the ASL dataset. (a) The JEPA encoder maintains a high-variance, non-collapsed distribution in PCA, reflecting an information-maximizing representation of hand kinematics. (b) While the supervised baseline achieves higher cluster cohesion in t-SNE, the corresponding PCA exhibits significant feature collapse, indicating the suppression of broader kinematic variance in favor of task-specific discriminative boundaries.

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